Gas & Combined Cycle Plant Cable Requirements - Consolidated Reference

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Purpose and scope

This document puts the cable requirements for a gas-fired / combined cycle power plant in one place: the consolidated requirements first, then a summary of every source behind them. It covers the plant modelled in this project: a 1x1 F-class CCGT with post-combustion carbon capture (CCS), a gas metering and LNG area, a modular gas-engine (RICE) plant and a 230 kV switchyard.

How to use it. Sections 2 to 6 are the working requirements: summary, deliverables, requirements by cable class, installation and testing. Sections 7 to 11 summarise each source. Section 12 lists what the owner, utility and vendors still have to supply.

Status of the sources. The sources were found by web search. The cloud environment's network policy blocked the source hosts (southwire.com, entergy.com, wbdg.org and others), so no source document was opened and read in full. Each summary states what the search results showed. Requirements marked *(practice)* come from US code and industry practice, not from a cited owner document. Confirm everything against the full documents before using it for procurement or construction.

Executive summary

A US gas or combined cycle plant should be cabled to NEC (NFPA 70), ICEA and IEEE practice. Use armored, tray-rated, FT4 flame-tested copper cable throughout, and Type MC-HL with listed fittings wherever an area is hazardous. The consolidated requirements are:

1. **Codes.** NEC 2023 for ampacity, raceway and hazardous areas. ICEA S-95-658 (600 V), S-93-639 (5-46 kV) and S-73-532 (control and instrument) for construction. IEEE 422 and 1185 for generating-station cable design and installation; IEEE 525 for the switchyard.
2. **Voltages.** 230 kV switchyard, 18-19 kV generator bus, 13.8 kV medium voltage (MV), 480 V low voltage (LV), 125 V DC, 120/208 V UPS.

3. **MV cable.** 15 kV MV-105, copper, 220 mil EPR at 133% insulation level, copper tape shield, PVC jacket. Armored (Armor-X type MC-HL) in plant areas. Nonsegregated phase bus above about 2,000 A.
4. **LV power and control cable.** 600 V XHHW-2 copper, Type MC-HL with continuous corrugated aluminum armor (CCW), or TC-ER where armor is not required. 3/C + ground for power. Control #14 AWG minimum, #10 AWG for CT circuits.
5. **VFD cable.** Symmetrical 3/C with three grounds and an overall shield; 360° shield termination at both ends. Armored (Armor-X VFD type) in tray; flexible TC-ER (Machine Flex type) for final drops to vibrating motors.
6. **Instrument cable.** Twisted shielded pairs, armored PLTC / ITC-HL. Intrinsically safe circuits segregated (blue). Type KX extension wire for thermocouples.
7. **Sizing.** 125% of load for motors and continuous loads, 115% for generators. Base ampacity from the 75 °C column, corrected for 40 °C ambient. Voltage drop ⩽ 3% per feeder and ⩽ 5% total.
8. **Hazardous areas.** Gas metering, LNG and fuel gas areas are Class I Div 1 or Div 2. Use MC-HL or ITC-HL with UL 2225 listed fittings, and seals in Div 1 (NEC 501.15).
9. **Testing.** Factory tests to ICEA. Insulation-resistance test on all cable. MV: VLF withstand plus tan δ to IEEE 400.2, acceptance to NETA ATS.
10. **Still owed.** The owner's design criteria, the transmission owner's switchyard standards, the hazardous area drawings and the vendor interface lists (section 12).

Project deliverables to date

Two deliverables exist, both on branch `claude/powerplant-example` of the `shardt87/ln` repository: a 3D line model of the plant and a cable schedule workbook at Rev D.

Deliverable	File	Content
3D line model	`examples/powerplant.go`	Gas metering / LNG, GT, HRSG and stack, ST hall, air-cooled condenser (ACC), CCS train, 8 containerized gas engines, step-up transformers, 4-bay 230 kV switchyard, 3 transmission towers. Equipment coordinates feed the cable lengths.
Cable schedule workbook	`docs/powerplant_cable_schedule.xlsx`	200 circuits, 152 equipment tags, 43 cable types, 17 raceways. Lengths, currents, loading, voltage drop, tray fill and BOQ are all formulas.
Workbook generator	`docs/make_cable_schedule.py`	Regenerates the workbook.

Workbook revisions:

Rev	Change
A	First issue, IEC conventions (11 kV / 400 V, mm², SWA armour).
B	Converted to US NEC practice: 13.8 kV / 480 V, AWG/kcmil, Type MC-HL (Armor-X type) cable, NEC ampacity and voltage drop.
C	CO_2 compressor VFD line and output circuits; ACC fan local disconnects; Machine Flex type flexible VFD final drops.

Rev	Change
D	3/C Armor-X MV-105 replaces single-conductor MV sets; nonsegregated phase bus from UAT-1; NEC 500 area classification; connector type and hub size at each cable end; pull and termination tracking register.

At Rev D, all 122 circuits with a current rating pass. Peak loading is 93%, peak voltage drop 2.7% and peak tray fill 60%. Cable diameters, masses and the 15 kV ampacities are still typical values: the manufacturer data sheets could not be downloaded.

Requirements by cable class

Each cable class has one construction, one sizing basis and one governing set of standards. The cable type codes match the workbook's Cable Types sheet.

Class	Construction	Sizing / rating basis	Governing standards	Workbook types
HV overhead (230 kV)	ACSR 795 kcmil "Drake", single or twin bundle; 7/16" EHS shield wire; OPGW 24 fibre	Ampacity at 40 °C ambient, 75 °C conductor; 3% sag allowance	IEEE 738, IEEE 525, transmission owner standards	ACSR-DRAKE, ACSR-DRAKE-2B, SW-7/16-EHS, OPGW-24
Generator bus (18-19 kV)	Isolated phase bus 10 kA (GT), 7 kA (ST), 3 kA taps	Generator rating, 1.0 factor	IEEE C37.23	IPB-10K, IPB-7K, IPB-3K
MV bus (13.8 kV)	Nonsegregated phase bus 3,000 A	Transformer secondary rating	IEEE C37.23	NSPB-3000
MV cable (13.8 kV)	15 kV MV-105, Cu, 220 mil EPR at 133%, Cu tape shield, PVC jacket, 3/C + ground, Armor-X type MC-HL	NEC 311.60, 90 °C column (MV-105 permits 105 °C); 125% for motors; #2 AWG minimum	ICEA S-93-639, UL 1072, NEC 311	MV-3C-2 to MV-3C-750

Class	Construction	Sizing / rating basis	Governing standards	Workbook types
LV power (480 V)	600 V XHHW-2, Cu, 3/C + ground, Armor-X type MC-HL (CCW aluminum armor, PVC jacket)	NEC 310.16 75 °C column × 0.88 for 40 °C; 125% continuous / motor; voltage drop ⩽ 3%	ICEA S-95-658, UL 1569, NEC 330, IEEE 1202 FT4	LV-3C-10 to LV-3C-500
LV busway	600 V Cu busway 2,500 / 4,000 A	Transformer secondary rating	UL 857	LVBD-2500, LVBD-4000
VFD, in tray	600 V 3/C + 3 symmetrical grounds, overall shield, Armor-X VFD type (SPEC 45226)	As LV power	ICEA S-95-658, NEC 430	LV-VFD-500
VFD, final drop	1000 V flexible tinned Cu, foil + braid shield, 3 symmetrical grounds, TPE jacket, TC-ER, NFPA 79 (Machine Flex type, SPEC 45443)	As LV power	NEC 336, NFPA 79	LV-MFX-2/0
DC and UPS	600 V XHHW-2 2/C or 1/C pair, MC-HL; circuit integrity (CI) cable for emergency lube oil pumps	125% load; voltage drop ⩽ 3%	NEC 310, NFPA 70 Ch. 9 Table 8	LV-2C-10, LV-2C-6, LV-2C-2, LV-1C-500
Control	600 V XHHW-2, Cu, #14 AWG minimum, #10 AWG for CT circuits, MC-HL	CT burden and coil voltage	ICEA S-73-532, IEEE 525	CTL-7C-12, CTL-12C-14, CTL-25C-14, CTL-12C-10

Class	Construction	Sizing / rating basis	Governing standards	Workbook types
Instrument	300 V twisted shielded pairs (individual + overall shield), #16-18 AWG, armored PLTC / ITC-HL; Type KX for thermocouples; RS-485 pair	Signal integrity; IS segregation	NEC 725, 727, 504; ICEA S-73-532	INS-1TSP-16 to INS-24TSP-18, INS-TC-12P-KX, INS-RS485
Fibre	Single-mode, 24 / 48 fibre, OFCR, interlocked aluminum armor	Redundant pairs on diverse routes	NEC 770	FO-24SM, FO-48SM
Grounding	Bare Cu 4/0 AWG grid; green insulated 2/0 AWG bonds; exothermic welds	Fault study	IEEE 80	EG-CU-4/0, EB-CU-2/0

(practice) rows: the sizing factors, voltage drop limits and minimum sizes come from US code and industry practice. They are not taken from an owner document.

Raceway, installation and hazardous areas

Cable runs on ladder tray above ground and in duct banks below it. Tray fill, segregation and hazardous-area wiring follow the NEC; installation limits follow IEEE 1185 and IEEE 422 *(practice)*.

Raceway

- Ladder tray: 600-1,000 mm wide, multi-tier. Power, control and instrument cables on separate tiers or trays.
- Fill: NEC 392.22. Cables 4/0 AWG and larger in a single layer. The workbook uses a simplified 40% area-fill check.
- Ampacity in tray: NEC 392.80(A). Duct banks: IEEE 835 / Neher-McGrath at earth thermal resistivity 90; the workbook uses a 0.70 correction factor.

- Switchyard control cable runs in a concrete trench to the relay building (IEEE 525).
- Large VFD cables need single-layer tray: the ACC fan deck uses two 600 mm risers.

Installation (IEEE 1185 / 422) *(practice)*

- Respect the cable maker's pulling tension, sidewall pressure and minimum bend radius.
- Slack allowance 10% and 3 m per termination in the workbook.
- Separate MV, LV, control and instrument cables; route redundant fibre on diverse paths.
- Fire-rated (circuit integrity) cable for emergency lube oil pumps and fire & gas loops.

Hazardous areas (NEC 500 / 501)

Area	Classification	Cable and fittings
LNG in-tank pumps, tank gauging	Class I Div 1	MC-HL or ITC-HL; UL 2225 Div 1 listed connectors; seals per NEC 501.15
Gas metering skid, regulators, fuel gas heater, BOG compressor, LNG area junction boxes	Class I Div 2	MC-HL or ITC-HL; Div 2 listed connectors
GT enclosure fans and F&G junction box	Class I Div 2	As Div 2
CCS amine and CO_2 areas	Unclassified (amine and CO_2 are not flammable)	Standard armored cable
Everything else	Unclassified	Watertight CCW armor connectors

These classifications are assumptions. Confirm them against the site hazardous area classification drawings. Intrinsically safe circuits stay segregated per NEC 504. Machine Flex type TC-ER cable is not permitted in Div 1.

Testing and commissioning

Every cable gets factory tests and a field insulation-resistance test. MV cable also gets a VLF withstand and tan δ test before energizing. The UFGS 26 05 13 reference list cites IEEE 400.2 (VLF), IEEE 404 (splices and terminations) and NETA ATS (acceptance

testing); the test values below are typical practice.

Test	Applies to	Requirement
Factory production tests	All cable	ICEA S-93-639 / S-95-658 / S-73-532 test reports with each reel.
Vertical tray flame test	All tray cable	IEEE 1202 / UL 1685 FT4.
Insulation resistance (IR)	600 V cable	1,000 V DC megger after pull and after termination *(practice)*.
Insulation resistance (IR)	15 kV cable	5,000 V DC megger *(practice)*.
VLF withstand + tan δ	15 kV cable, splices, terminations	IEEE 400.2; acceptance per NETA ATS.
Shield and armor continuity	Shielded and armored cable	Continuity and bonding at both ends *(practice)*.
Continuity and point-to-point	Control and instrument	Checked against wiring diagrams *(practice)*.
Fibre	All fibre	OTDR and insertion loss *(practice)*.
Ground grid	Grounding	Fall-of-potential test of grid resistance to IEEE 81 *(practice)*.

The workbook's Termination & Pull sheet has columns to record reel number, pull date, termination date at each end, IR result and VLF date per cable. Its status rolls up to Not started, In progress or Complete.

For in-service condition monitoring, the EPRI medium-voltage cable guides (section 9) describe tan δ, partial discharge and dielectric spectroscopy.

Sources: owner and utility documents

The closest public match to an owner's cable requirement is Entergy's combined-cycle RFP scope book. Its cable clauses were not visible in the search results, so it is the first document to read in full.

Document	What it is	Cable-relevant content seen
Entergy 2024 CCCT RFP, Appendix B-4 B-O-T Scope Book	The utility's scope and design criteria for a build-own-transfer combined-cycle plant, including Attachment A-4, Project Requirements and Design Criteria.	Telecom and interface cabling scope only; MV, LV and raceway clauses not reached.
Entergy 2026 Model CCCT Scope Book (draft)	Updated model scope book, under Entergy Arkansas review.	Telecom and security cabling only.
Entergy Arkansas Scope Book Redline	2025 revision of the scope book.	Not reached.
Entergy Solar BOT Scope Book	Same utility, solar plant.	XLPE-insulated MV AC cable; 600 V cable in conduit.
Entergy Wind BOT Scope Book	Same utility, wind plant.	Halogen-free, fire-retardant, self-extinguishing cable.
Entergy BESS BOT Scope Book	Same utility, battery storage.	MV switchgear, auxiliary transformers, AC/DC supplies and all associated cables in contractor scope.
Fichtner design basis report, 355 MW CCPP (Kondapalli, India)	Electrical design basis for a combined-cycle plant.	Covers cables, busbars, protection, grounding, layout and codes; IEC-based. Hosted on Scribd.

The other-technology scope books show the same utility's preferences (XLPE MV cable, low-halogen fire-retardant cable). The combined-cycle book may differ.

Sources: government guide specifications

The US federal guide specifications are complete, editable cable specs. They are the quickest base for a project cable specification.

Document	Scope	Cable-relevant content seen
UFGS 26 05 13, Medium-Voltage Cables (Aug 2025)	Shielded and nonshielded MV power cable, portable cable, splices and terminations, fireproofing in manholes and tunnels. Supersedes UFGS 26 05 13.00 10.	References IEEE 400.2 (VLF field testing), IEEE 404 and NETA ATS.
UFGS 26 05 13.00 10 (Nov 2023, superseded)	Previous edition.	For projects tied to the older edition.
UFGS 26 05 19.00 10, Insulated Wire and Cable (Feb 2026)	600 V and below wire and cable.	Test reports for cable data, high-voltage test source, shielded cable above 2 kV, flame tests, installation and independent tests.
VA 26 05 13, Medium-Voltage Cables	VA guide spec.	EPR insulation at 133%; semiconducting insulation shield; helical copper tape shield; heavy-duty PVC jacket.
VA 26 05 19, Low-Voltage Conductors and Cables	VA guide spec, 600 V and below.	Furnishing, installation, connection and testing of LV conductors.
University of Washington MV wire, cable and terminations standard	Owner standard.	133% insulation level: 115 mil at 5 kV, 220 mil at 15 kV.

These guide specs match the workbook's MV construction: EPR at 133% (220 mil at 15 kV), copper tape shield and PVC jacket. The UFGS VLF and tan δ acceptance values were not visible and need reading in full.

Sources: codes and industry standards

Owner specifications cite these standards, and most are paid documents. IEEE 422 and IEEE 525 are listed as inactive and need a status check with IEEE.

Standard	Covers	Used for
NFPA 70 (NEC) 2023	310.16 / 311.60 ampacity; 330 Type MC; 336 Type TC; 392 cable tray; 430 motors; 445 generators; 500-504 hazardous areas and IS; 725 PLTC; 727 ITC; 770 fibre; Chapter 9 Tables 8-9	Sizing, raceway, hazardous areas, voltage drop
ICEA S-95-658 (NEMA WC70)	600 V power cable	LV construction
ICEA S-93-639 (NEMA WC74)	5-46 kV shielded power cable	MV construction
ICEA S-73-532	Control, thermocouple and instrument cable	Control and instrument construction
UL 1569 / UL 2225 / UL 1072 / UL 1277	MC cable; cables and fittings for hazardous locations (MC-HL); MV cable; tray cable	Listings
IEEE 1202 / UL 1685	Vertical tray flame test (FT4)	All tray cable
IEEE 422 (1986, inactive)	Design and installation of wire and cable systems in generating stations, nuclear and non-nuclear; excludes switchyards	Plant cable design basis
IEEE 1185 (2019, active)	Cable installation in generating stations and industrial facilities: tray, conduit, duct bank; avoiding damage in pulling and testing	Installation practice
IEEE 525 (1992 / 2016)	Cable systems in substations	Switchyard cable
IEEE 690 (1984)	Class 1E nuclear cable	Not applicable here
IEEE 835	Power cable ampacity tables	Duct bank and tray ampacity

Standard	Covers	Used for
IEEE 400.2 / IEEE 404 / NETA ATS	VLF field testing; splices and terminations; acceptance testing	Testing
IEEE 738 / IEEE C37.23 / IEEE 80	Bare conductor ampacity; metal-enclosed bus; substation grounding	HV, busway, grounding
EPRI MV cable guides: 1013187, 1022968, 1021070	MV cable life-cycle and aging management; EPR wet aging; tan δ and partial discharge testing	Condition monitoring, long-term reliability

Sources: cable product data

Southwire's Armor-X and Machine Flex lines are named as examples of the construction; equivalent products from other makers fit the same requirements. Search results confirmed construction, temperature and hazardous-location ratings. Diameters, masses and size ranges were not readable.

Product (Southwire spec)	Construction	Ratings seen
SPEC 45352: Armor-X 600 V power cable with ground	Cu, XLPE XHHW-2, continuous corrugated welded armor, PVC jacket; Type MC-HL	90 °C wet/dry; 130 °C emergency; 250 °C short circuit; -50 °C cold bend; ICEA S-95-658, S-73-532; IEEE 1202 FT4; Class I-III Div 1 & 2
SPEC 45227: Armor-X 600 V control cable with ground	7-strand tinned Cu, 30 mil XLPE, Armor-X, PVC jacket	As SPEC 45352
SPEC 45226: Armor-X VFD cable with three grounds	3/C or 4/C XHHW-2, three symmetrical grounds, Armor-X, PVC jacket	600 V
SPEC 45302: Armor-X VFD cable, LSZH jacket	As 45226 with low-smoke zero-halogen jacket	600/1000 V

Product (Southwire spec)	Construction	Ratings seen
SPEC 46454: 3/C 15 kV Armor-X MV-105	3/C Cu, 220 mil NL-EPR at 133%, tape shield, Armor-X, PVC jacket	105 °C normal; 140 °C emergency; 250 °C short circuit; UL 1072; ICEA S-93-639; Class I-III Div 1 & 2
SPEC 45443: Machine Flex TC-ER VFD, three symmetrical grounds	Flexible tinned Cu, XLPE, three tinned grounds, shield, TPE jacket	1000 V; TC-ER; NFPA 79; -40 °C cold bend; FT4
SPEC 45442: reduced-diameter flexible VFD	Flexible tinned Cu, XHHW-2, foil + braid shield, TPE jacket	600/1000 V; TC-ER; 90 °C wet/dry
SPEC 45440: Machine Flex RHH/RHW-2	RHH/RHW-2, TPE jacket	TC-ER; sunlight resistant; direct burial; NFPA 79
SPEC 45255: TC-ER EPR/CPE 600 V	EPR insulation, CPE jacket, unarmored tray cable	90 °C; UL 1277; ICEA S-95-658; Class I-II Div 2

One gap remains: the Machine Flex size range was not confirmed. A 2 AWG 3/C version exists; the workbook's 2/0 AWG size still needs checking.

Sources: plant arrangement references

These public permit, study and design documents show typical plant layouts and electrical arrangements. They inform equipment positions and the one-line diagram, not cable construction. The 3D model and workbook use representative values, not data from these projects.

Area	Document	Relevance
CCPP + LNG + gas metering	Shannon Technology & Energy Park, list of planning drawings	One application with LNG terminal, CCPP and separate gas metering / regulating enclosures
CCPP layout	Shannon CCPP general arrangement	General arrangement drawing
CCPP layout	Tees CCPP indicative generating station, sheet 1	UK planning-stage layout

Area	Document	Relevance
CCPP layout	ESC Brooke County CCGT air permit, WV	Conceptual plant layout (Appendix A)
CCS	Bechtel / Panda Sherman NGCC carbon capture FEED, final report (OSTI)	2x2x1 758 MW retrofit; capture plant scope
CCS	NETL: EPRI CalCapture (Elk Hills) FEED	Deliverables include a plot plan
CCS	NETL: Calpine Deer Park FEED	Commercial-scale modular capture
Gas metering	Adams County CO regulator station site plan	Regulator station layout
RICE	KYMEA Energy Center (4 x Wärtsilä 18V50SG), KY PSC 2024-00290	About 75 MW engine plant
RICE	We Energies Paris RICE project	7 x 19 MW engines
Switchyard / one-line	PJM facilities study, 1,100 MW combined cycle	Substation one-lines
Switchyard / one-line	SPP GEN-2008-006 facility study	2-on-1 CCGT, 138 kV one-lines, a step-up transformer per unit
Schedule format	CFIHOS cable schedule register, GlobalSpec CR4, AVEVA Electrical, Engineers Universe	Standard cable schedule columns: tag, type, cores, size, voltage, from/to, gland at each end, length, routing drawing, tracking

Gaps: what the owner, utility and vendors must provide

These inputs fix the cable package more than any public source. Until they arrive, the related workbook values stay typical.

- [ ] **Owner design criteria and cable specification:** EPR or XLPE insulation; MC-HL or TC-ER; jacket (PVC or LSZH); minimum sizes; spare conductors; tray fill and segregation policy.

- [ ] **Transmission owner standards:** switchyard control cable type, CT and VT wiring, relay building layout, fibre and OPGW requirements, interconnection one-line.
- [ ] **Hazardous area classification drawings** for gas metering, LNG, fuel gas, BOG compressor and GT enclosure (replaces the assumed Div 1 / Div 2 areas).
- [ ] **Vendor interface lists** for GT, ST, HRSG, CCS, CO_2 compressor, VFDs and gas engines: vendor-supplied versus field cable, terminal box details, VFD output cable requirements.
- [ ] **Load list** with motor ratings, VFD versus direct-on-line starting, and duty, to replace the representative loads.
- [ ] **Short-circuit study and protection settings:** cable short-circuit withstand (ICEA P-32-382) and motor-starting voltage dip, especially the 6.5 MW booster fan.
- [ ] **Manufacturer data sheets** for the chosen cable: diameter, mass, reel length and size range (Machine Flex 2/0 AWG to confirm).
- [ ] **Test and commissioning specification:** NETA ATS scope, IR test voltages, VLF and tan δ acceptance values.
- [ ] **Full reads of the Entergy CCCT scope book and UFGS 26 05 13 / 26 05 19.** Their hosts are blocked in this environment; allow `rfp.entergy.com`, `entergyarkansas.com`, `wbdg.org` and `southwire.com` in the network settings to read them.

Sources

All sources were found by web search on 2026-10-18 and none was opened in full (section 1). Links are also given in sections 7-11.

- **Owner / utility:** Entergy 2024 CCCT Scope Book · Entergy 2026 Model CCCT Scope Book · Entergy Arkansas Redline · Entergy Solar · Entergy Wind · Entergy BESS · Fichtner CCPP design basis
- **Guide specifications:** UFGS 26 05 13 · UFGS 26 05 13.00 10 · UFGS 26 05 19.00 10 · VA 26 05 13 · VA 26 05 19 · UW MV cable standard
- **Standards and research:** IEEE 422 · IEEE 1185 · IEEE 525 · EPRI 1013187 · EPRI 1022968
- **Cable products (Southwire):** 45352 · 45227 · 45226 · 45302 · 46454 · 45443 · 45442 · 45440 · 45255
- **Plant arrangement and schedule format:** see the table in section 11.